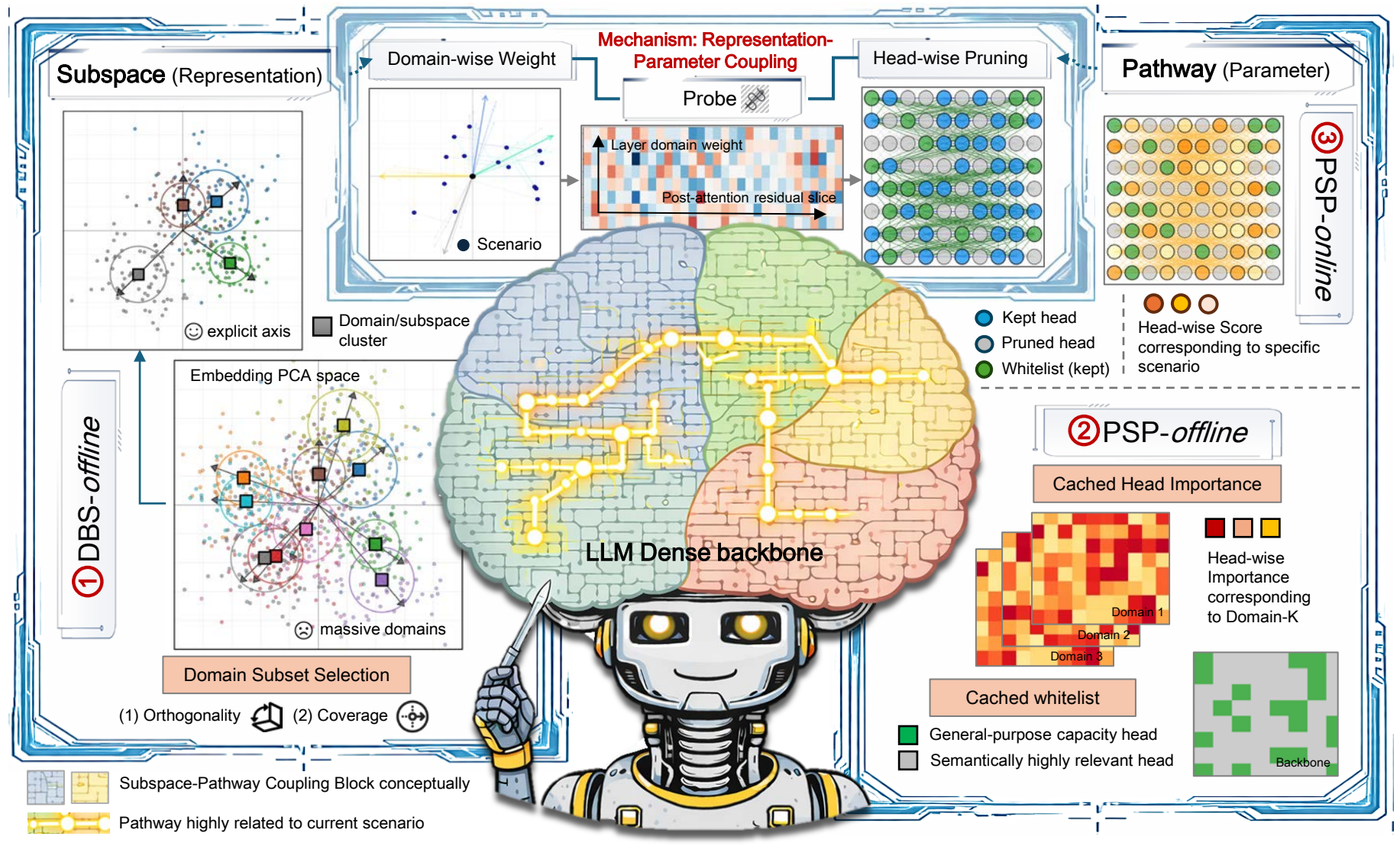
Subspace (Representation)
Domain-wise Weight
Mechanism: Representation-Parameter Coupling
Head-wise Pruning
Pathway (Parameter)
Probe
Scenario
Layer domain weight
Post-attention residual slice
explicit axis
Domain/subspace cluster
Embedding PCA space
massive domains
① DBS-offline
Domain Subset Selection
(1) Orthogonality
(2) Coverage
LLM Dense backbone
Kept head
Pruned head
Whitelist (kept)
Head-wise Score corresponding to specific scenario
③ PSP-online
② PSP-offline
Cached Head Importance
Head-wise Importance corresponding to Domain-K
Domain 1
Domain 2
Domain 3
Cached whitelist
General-purpose capacity head
Semantically highly relevant head
Backbone
Subspace-Pathway Coupling Block conceptually
Pathway highly related to current scenario